Student: Xirong Zhang **Date:** 11/30/25	**Instructor:** Dmitriy Ivanov **Course:** Math 120 Fall 2025	**Assignment:** Section 2.3

1. Determine if the matrix below is invertible. Use as few calculations as possible. Justify your answer.

$$\begin{bmatrix} 5 & 0 & 0 \\ -2 & -3 & 0 \\ 5 & 7 & -4 \end{bmatrix}$$

Choose the correct answer below.

- **A.** The matrix is invertible. The given matrix has three pivot positions.
- **B.** The matrix is not invertible. If the given matrix is A, the equation A**x** = **b** has no solution for some **b** in $\mathbb{R}^3$.
- **C.** The matrix is not invertible. The given matrix has two pivot positions.
- **D.** The matrix is invertible. If the given matrix is A, there is a 3×3 matrix C such that $CI = A$.

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2. Determine if the matrix below is invertible. Use as few calculations as possible. Justify your answer.

$$\begin{bmatrix} 4 & 0 & -4 \\ 3 & 0 & 5 \\ -5 & 0 & 9 \end{bmatrix}$$

Choose the correct answer below.

- **A.** The matrix is not invertible. If the given matrix is A, the columns of A do not form a linearly independent set.
- **B.** The matrix is invertible. The given matrix has 2 pivot positions.
- **C.** The matrix is not invertible. If the given matrix is A, the equation A**x** = **0** has only the trivial solution.
- **D.** The matrix is invertible. The columns of the given matrix span $\mathbb{R}^3$.

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3. Determine if the matrix below is invertible. Use as few calculations as possible. Justify your answer.

$$\begin{bmatrix} 0 & 4 & -4 \\ 1 & 0 & 3 \\ -3 & -8 & -1 \end{bmatrix}$$

Choose the correct answer below.

- **A.** The matrix is invertible. The columns of the given matrix span are linearly dependent.
- **B.** The matrix is not invertible. If the given matrix is A, A is not row equivalent to the $n \times n$ identity matrix.
- **C.** The matrix is not invertible. If the given matrix is A, the equation A**x** = **b** has a solution for at least one **b** in $\mathbb{R}^3$.
- **D.** The matrix is invertible. The given matrix has 3 pivot positions.

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4. Determine if the matrix below is invertible. Use as few calculations as possible. Justify your answer.

$$\begin{bmatrix} 1 & -2 & -5 \\ 0 & 4 & 2 \\ -2 & 1 & 0 \end{bmatrix}$$

Determine if the matrix below is invertible. Use as few calculations as possible. Justify your answer.

- **A.** The matrix is not invertible. In the given matrix the columns do not form a linearly independent set.
- **B.** The matrix is invertible. The given matrix is not row equivalent to the $n \times n$ identity matrix.
- ✔ **C.** The matrix is invertible. The given matrix has 3 pivot positions.
- **D.** The matrix is not invertible. If the given matrix is A, the equation $A\mathbf{x} = \mathbf{b}$ has no solution for at least one $\mathbf{b}$ in $\mathbb{R}^3$.

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5. For this exercise assume that the matrices are all $n \times n$. The statement in this exercise is an implication of the form "If "statement 1", then "statement 2"." Mark an implication as True if the truth of "statement 2" always follows whenever "statement 1" happens to be true. Mark the implication as False if "statement 2" is false but "statement 1" is true. Justify your answer.

If there is an $n \times n$ matrix D such that $AD = I$, then there is also an $n \times n$ matrix C such that $CA = I$.

Choose the correct answer below.

- **A.** The statement is false. It does not follow from the Invertible Matrix Theorem that if $AD = I$, then $CA = I$.
- **B.** The statement is true. By the Invertible Matrix Theorem, if $AD = I$ then A, D, or both is/are the identity matrix. Therefore, $CA = I$.
- ✔ **C.** The statement is true. By the Invertible Matrix Theorem, if there is an $n \times n$ matrix D such that $AD = I$, then it must be true that there is also an $n \times n$ matrix C such that $CA = I$.
- **D.** The statement is false. Matrix multiplication is not commutative. It is possible that only D (and not A) is invertible in the equation $AD = I$. This implies that $CA = I$ is only true when C is invertible (for cases where A is not invertible), but it is not given that C is invertible.

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6. For this exercise assume that the matrices are all $n \times n$. The statement in this exercise is an implication of the form "If "statement 1", then "statement 2"." Mark an implication as True if the truth of "statement 2" always follows whenever "statement 1" happens to be true. Mark the implication as False if "statement 2" is false but "statement 1" is true. Justify your answer.

If the columns of A span $\mathbb{R}^n$, then the columns are linearly independent.

Choose the correct answer below.

- **A.** The statement is false. The Invertible Matrix Theorem states that if the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ into $\mathbb{R}^n$, then A is not invertible. Therefore, the columns are linearly dependent.
- **B.** The statement is true. The Invertible Matrix Theorem states that if the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ does not map $\mathbb{R}^n$ into $\mathbb{R}^n$, then A is invertible. Therefore, the columns are linearly independent.
- ✔ **C.** The statement is true. The Invertible Matrix Theorem states that if the columns of A span $\mathbb{R}^n$, then matrix A is invertible. Therefore, the columns are linearly independent.
- **D.** The statement is false. The Invertible Matrix Theorem states that if the columns of A span $\mathbb{R}^n$, then matrix A is not invertible. Therefore, the columns are linearly dependent.

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7. For this exercise assume that the matrices are all $n \times n$. The statement in this exercise is an implication of the form "If "statement 1", then "statement 2"." Mark an implication as True if the truth of "statement 2" always follows whenever "statement 1" happens to be true. Mark the implication as False if "statement 2" is false but "statement 1" is true. Justify your answer.

If the equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution, then A has fewer than n pivot positions.

Choose the correct answer below.

- **A.** The statement is false. By the Invertible Matrix Theorem, if the equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution, then the columns of A do not form a linearly independent set. Therefore, A has n pivot positions.
- **B.** The statement is true. By the Invertible Matrix Theorem, if the equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution, then matrix A is not invertible. Therefore, A has fewer than n pivot positions.
- **C.** The statement is false. By the Invertible Matrix Theorem, if the equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution, then matrix A is invertible. Therefore, A has n pivot positions.
- **D.** The statement is true. By the Invertible Matrix Theorem, if the equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution, then the columns of A form a linearly independent set. Therefore, A has fewer than n pivot positions.

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8. For this exercise assume that the matrices are all $n \times n$. The statement in this exercise is an implication of the form "If "statement 1", then "statement 2"." Mark an implication as True if the truth of "statement 2" always follows whenever "statement 1" happens to be true. Mark the implication as False if "statement 2" is false but "statement 1" is true. Justify your answer.

If A^T is not invertible, then A is not invertible.

Choose the correct answer below.

- **A.** The statement is true. By the Invertible Matrix Theorem, if A^T is not invertible, then there is an $n \times n$ matrix C such that $CA = I$. This means that A must not be invertible.
- **B.** The statement is false. By the Invertible Matrix Theorem, if A^T is not invertible, then there is not an $n \times n$ matrix C such that $CA = I$. Therefore, A is invertible.
- **C.** The statement is false. By the Invertible Matrix Theorem, if A^T is not invertible, then all statements in the theorem are true, including A is invertible. Therefore, A is invertible.
- **D.** The statement is true. By the Invertible Matrix Theorem, if A^T is not invertible, then all statements in the theorem are false, including A is invertible. Therefore, A is not invertible.

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9. Is it possible for a 5×5 matrix to be invertible when its columns do not span $\mathbb{R}^5$? Why or why not?

Select the correct choice below.

- **A.** It will depend on the values in the matrix. According to the Invertible Matrix Theorem, a square matrix is only invertible if it is row equivalent to the identity.
- **B.** It is possible; according to the Invertible Matrix Theorem all square matrices are always invertible.
- **C.** It is not possible; according to the Invertible Matrix Theorem an $n \times n$ matrix cannot be invertible when its columns do not span $\mathbb{R}^n$.
- **D.** It is possible; according to the Invertible Matrix Theorem an $n \times n$ matrix can be invertible when its columns do not span $\mathbb{R}^n$.

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10. If L is $n \times n$ and the equation L**x** = **0** has the trivial solution, do the columns of L span $\mathbb{R}^n$? Why?

Select the correct choice below.

- **A.** This fact gives no information about the columns of L. The equation L**x** = **0** always has the trivial solution.
- **B.** The columns of L do not span $\mathbb{R}^n$. If L**x** = **0** has the trivial solution, then according to the Invertible Matrix Theorem, L is not invertible and the columns of L do not span $\mathbb{R}^n$.
- **C.** If L**x** = **0** has the trivial solution and L is square, then L is invertible, and the columns of L span $\mathbb{R}^n$.
- **D.** If L**x** = **0** has the trivial solution, then according to the Invertible Matrix Theorem, L**x** = **b** has a solution for all **b** and the columns of L span $\mathbb{R}^n$.